# Year 9
# Mathematics

## Unit 3: Sequences and Algebra

3.1 Arithmetic and Quadratic Sequences

# Arithmetic and Quadratic Sequences

STARTER · 5 MIN

**Date:** _____ **Title: Arithmetic and Quadratic Sequences**

**LEARNING OBJECTIVES**

- Use the nth term to write a linear or quadratic sequence
- Find the nth term of an arithmetic sequence
- Decide if a number is a term in a sequence

**STARTER** *Answer in your book. No calculators.*

1. Work out $3 \times 5 - 2$
2. Work out $4^2$
3. Work out $2 \times 3^2$
4. Next term: 5, 9, 13, …

# Starter — Answers

STARTER

*Tick or correct in green pen*

1. **13**

2. **16**

3. **18**

4. **17**

# Success Criteria

SUCCESS CRITERIA

*By the end of this lesson:*

| | | |
|---|---|---|
| BRONZE | I can use the nth term to write a linear or quadratic sequence | ☐ |
| SILVER | I can find the nth term of an arithmetic sequence | ☐ |
| GOLD | I can decide if a number is a term in a sequence | ☐ |

*Tick each box in the plenary when you can do it on your own.*

# Introduction — Arithmetic and Quadratic Sequences

INTRODUCTION · 3 MIN

The nth term is a rule that gives any term from its position n.

## Linear (arithmetic)

The terms go up by the same amount.

3n + 2:  5, 8, 11, 14, …

## Quadratic

Has $n^2$. The second differences are equal.

$n^2$:  1, 4, 9, 16, …

## Substitute

Put n = 1, 2, 3, … into the rule.

$3n^2$ when n = 2:  3 × 4 = 12

## nth term

Difference × n, then adjust.

5, 8, 11, …:  3n + 2

# Examples 1 and 2 — Generating Sequences

EXAMPLES · 5 MIN

BRONZE

*Watch and copy into your book*

**Hint** put n = 1, 2, 3, 4  $3n^2 = 3 \times n^2$

## Example 1

Write down the first four terms of the sequence with nth term $5n + 2$.

## Example 2

Write down the first four terms of each sequence.

(a) $2n^2$  (b) $n^2 + 3$

# Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

*Check your copy and correct in green pen*

## Example 1

**Put n = 1, 2, 3, 4**

$5 \times 1 + 2 = 7$, $5 \times 2 + 2 = 12$

$5 \times 3 + 2 = 17$, $5 \times 4 + 2 = 22$

**7, 12, 17, 22**

Tip: the terms go up by 5 each time.

## Example 2

**Square n first**

(a) $2 \times 1$, $2 \times 4$, $2 \times 9$, $2 \times 16$

$= 2, 8, 18, 32$

**(b) 4, 7, 12, 19**

Tip: $2n^2$ means $2 \times n^2$, not $(2n)^2$.

# Task 1

PRACTICE · 9 MIN

BRONZE *Write down the first four terms*

**Hint** put n = 1, 2, 3, 4 $3n^2 = 3 \times n^2$

1. $3n + 1$
2. $6n - 4$
3. $7n - 3$
4. $n^2$
5. $3n^2$

★ $n^2 + 5$

# Task 1 — Answers

CORRECTIONS

BRONZE

*Tick or correct in green pen*

1. 4, 7, 10, 13
2. 2, 8, 14, 20
3. 4, 11, 18, 25
4. 1, 4, 9, 16
5. 3, 12, 27, 48

★ 6, 9, 14, 21

# Examples 3 and 4 — Finding the nth Term

EXAMPLES · 5 MIN

SILVER

*Watch and copy into your book*

**Hint** nth term = difference × n, then adjust
equal 2nd differences: quadratic

## Example 3

Find the nth term of

5, 8, 11, 14, …

## Example 4

Is each sequence linear or quadratic?

(a) 4, 9, 16, 25, …

(b) 7, 11, 15, 19, …

# Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

*Check your copy and correct in green pen*

## Example 3

**The difference is 3: so 3n**

3n:  3, 6, 9, 12

each term is 2 more

**nth term = 3n + 2**

Tip: check: n = 4 gives 14.

## Example 4

**Look at the differences**

(a)  +5, +7, +9

2nd differences +2: quadratic

**(b)  +4 each time: linear**

Tip: equal second differences: quadratic.

# Task 2

PRACTICE · 9 MIN

SILVER

*Questions 1–4: find the nth term. 5–6: linear or quadratic?*

**Hint** nth term = difference × n, then adjust
equal 2nd differences: quadratic

1. 4, 7, 10, 13, …
2. 2, 8, 14, 20, …
3. 11, 15, 19, 23, …
4. 1, 6, 11, 16, …
5. 3, 6, 11, 18, 27, …

★ 2, 9, 16, 23, …

# Task 2 — Answers

CORRECTIONS

SILVER

*Tick or correct in green pen*

1. $3n + 1$
2. $6n - 4$
3. $4n + 7$
4. $5n - 4$
5. quadratic

★ linear

# Examples 5 and 6 — Is It a Term?

EXAMPLES · 5 MIN

GOLD

*Exam-style: show all your working*

**Hint** set the nth term = the number
whole number n: it is a term

## Example 5

(2 marks)

Is 85 a term in the sequence with nth term 4n + 1?

Explain your answer.

## Example 6

(3 marks)

The nth term is $n^2 - 3$.

(a) Find the 10th term.

(b) Which term is 141?

# Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

*Check your copy and correct in green pen*

## Example 5

**Set the nth term equal to 85**

$4n + 1 = 85$

$4n = 84,\ \ n = 21$

**Yes: it is the 21st term**

Tip: a whole number n means it is a term.

## Example 6

**(a) n = 10  (b) solve**

(a) $10^2 - 3 = 97$

(b) $n^2 - 3 = 141,\ \ n^2 = 144$

**n = 12: the 12th term**

Tip: n is a position, so it is positive.

# Task 3

PRACTICE · 9 MIN

GOLD *Show your working*

**Hint** set the nth term = the number whole number n: it is a term

1. Is 50 a term of $3n - 1$?

2. Is 75 a term of $4n + 5$?

3. Is 102 a term of $7n + 4$?

4. Find the 8th term of $n^2 + 6$

5. Which term of $n^2 - 5$ is 139?

★ Which term of $2n^2$ is 288?

# Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1 **Yes: n = 17**

2 **No: n = 17.5**

3 **Yes: n = 14**

4 **70**

5 **12th term**

★ **12th term**

# Plenary — Exit Ticket

PLENARY · 5 MIN

*Answer in your book. Try all three.*

**BRONZE** Write the first three terms of 2n + 5

**SILVER** Find the nth term of 6, 10, 14, 18, …

**GOLD** Is 60 a term of 4n + 2? Explain.

**Reflect:** Look back at the learning objectives. Which ones can you do now?

# Plenary — Answers

PLENARY

*Tick or correct in green pen*

BRONZE **7, 9, 11**

SILVER **4n + 2**

GOLD **No: 4n = 58, n = 14.5**

# Exam Practice — Generating Sequences

EXAM PRACTICE · 10 MIN

BRONZE

*IGCSE exam-style: show all your working*

**Hint** put n = 1, 2, 3, 4  $3n^2 = 3 \times n^2$

**Question 1** (1 mark)

Write down the first four terms of the sequence with nth term $6n - 5$.

.....

**Question 2** (2 marks)

Write down the first four terms of the sequence with nth term $2n^2 + 1$.

.....

# Exam Practice — nth Term

EXAM PRACTICE · 10 MIN

SILVER

*IGCSE exam-style: show all your working*

**Hint** nth term = difference × n, then adjust
equal 2nd differences: quadratic

## Question 3

(2 marks)

Find the nth term of the sequence
9, 13, 17, 21, …

.....

## Question 4

(2 marks)

Which of these sequences is linear and which is quadratic?

**(a)** $3n + 4$ (1)

.....

**(b)** $2n^2 - 1$ (1)

.....

# Exam Practice — Is It a Term?

EXAM PRACTICE · 10 MIN

GOLD

*IGCSE exam-style: show all your working*

**Hint** set the nth term = the number
whole number n: it is a term

## Question 5

(2 marks)

Is 120 a term in the sequence with nth term 5n − 3?
Give a reason for your answer.

.....

## Question 6

(3 marks)

A sequence has nth term $3n^2 + 1$.

**(a)** Find the 5th term. (1)

.....

**(b)** Which term is 301? (2)

.....

# Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

*Mark your work: M = method mark, A = accuracy mark, B = independent mark*

## Question 1

**1, 7, 13, 19** B1

Tip: put n = 1, 2, 3, 4

## Question 2

**3, 9, 19, 33** B2

Tip: square n, double it, then add 1.

# Exam Practice — Mark Scheme

CORRECTIONS

SILVER

*Mark your work: M = method mark, A = accuracy mark, B = independent mark*

## Question 3

difference 4: 4n — M1

**4n + 5** — A1

Tip: 4n gives 4, 8, 12: add 5

## Question 4

**(a) linear** — B1

**(b) quadratic** — B1

Tip: $n^2$ in the rule: quadratic.

# Exam Practice — Mark Scheme

CORRECTIONS

GOLD

*Mark your work: M = method mark, A = accuracy mark, B = independent mark*

## Question 5

$5n - 3 = 120$, $5n = 123$ — M1

**$n = 24.6$: not whole, so no** — A1

Tip: n must be a whole number.

## Question 6

**(a) $3 \times 25 + 1 = 76$** — B1

(b) $3n^2 = 300$, $n^2 = 100$ — M1

**$n = 10$: the 10th term** — A1

Tip: undo + 1, then ÷ 3, then square root